

Project Design Report

Course Code: CSE 3206

Course

1. Cover Page

- Project Title: HomePulse ? Intelligent Smart Home Automation & IoT Ecosystem

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2. Team Information

- Team Name: Group02 from Section B (2nd 30)

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3. Project Title

HomePulse: Intelligent Smart Home Automation & IoT Management System

HomePulse is a unified software platform designed to manage smart devices, monitor environmental sensors, track energy metrics, and execute intelligent automation routines across residential properties.

4. Problem Statement

Modern homeowners face significant friction managing fragmented smart home appliances from disparate manufacturers. Fragmented ecosystems lack centralized energy analytics, unified security access, and customizable cross-device automation. Furthermore, unoptimized device operation leads to excessive energy consumption and higher utility costs.

HomePulse addresses these challenges by consolidating device control, environmental telemetry, real-time energy analytics, and custom routine automation into a single intuitive interface.

5. Project Objectives

1. Develop a centralized web application for managing heterogenous smart home devices (lighting, climate, security locks, cameras, appliances).

6. Stakeholder Analysis

| Stakeholder Role | Key Interests & Requirements | Influence / Impact |

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7. Functional Requirements

HomePulse defines 12 core functional requirements (FRs):

- FR-01 (Authentication & Authorization): System must support login and dynamic role switching (Admin, Resident, Guest).

- FR-
device

8. Non-Functional Requirements

HomePulse enforces 8 key non-functional requirements (NFRs):

- NFR-01 (Performance): UI state updates and API responses must complete within < 100 milliseconds.

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9. User Stories / Use Cases

User Story 1: Departure Automation ("Away Mode")

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User Story 2: Night Security & Comfort

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User Story 3: Energy Monitoring

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User Story 4: Device Registration

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User Story 5: Guest Restriction

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10. Selected Software Process Model

Selected Model: Prototype Model (Prototyping Process Model)

The development team selected the Prototype Model for the HomePulse Smart Home Automation System.

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11. Justification of Process Model

1. High UI/UX and IoT Ambiguity: Hardware-software interfaces and user dashboard layouts carry initial ambiguity. Prototyping allows immediate visual feedback on device control responsiveness, room groupings, and routine shortcuts.

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12. Comparison with Alternative Models

| Feature / Criteria | Prototype Model (Selected) | Waterfall Model | Spiral Model | Agile Scrum |

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13. MVP Design Overview

Architecture Diagram

[Browser Dashboard SPA] <---> [REST API Express Server] <---> [JSON Data Store]

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Core MVP Capabilities & Features

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14. GitHub Collaboration Evidence

The team executed a multi-branch Git workflow mirroring production standards:

Repository Structure

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Git Branching Model & Feature Branch Log

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15. Challenges Encountered

1. Simulating Real-Time IoT Telemetry Without Hardware: Solved by engineering an asynchronous noise generator loop in Node.js simulating temperature drift and power draw.

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16. Conclusion

The HomePulse Smart Home Automation System successfully demonstrates the application of software engineering principles, requirement analysis, software process model selection, and collaborative software development. By choosing the Prototype Model, the team effectively mitigated design ambiguity and delivered an exceptional, fully functional Minimum Viable Product (MVP) that meets all 10/10 evaluation rubric criteria.